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EXPERIMENT 3

Exp No:3.a Conduct an experiment to show data visualization using line plot

AIM:

To conduct an experiment to show data visualization using line plot

DESCRIPTION:

Take any sample data either through csv file or data fetched directly through code.

CODE:

```
import matplotlib.pyplot as sales

Months = ["Jan","Feb","Mar","Apr","May","Jun"]

Product_A = [120,150,180,160,200,240]

Product_B = [100,130,160,170,190,210]

sales.plot(Months,Product_A,color="green",label="PRODUCT A")

sales.plot(Months,Product_B,color="red",label="PRODUCT B")

sales.title("MONTHLY SALES")

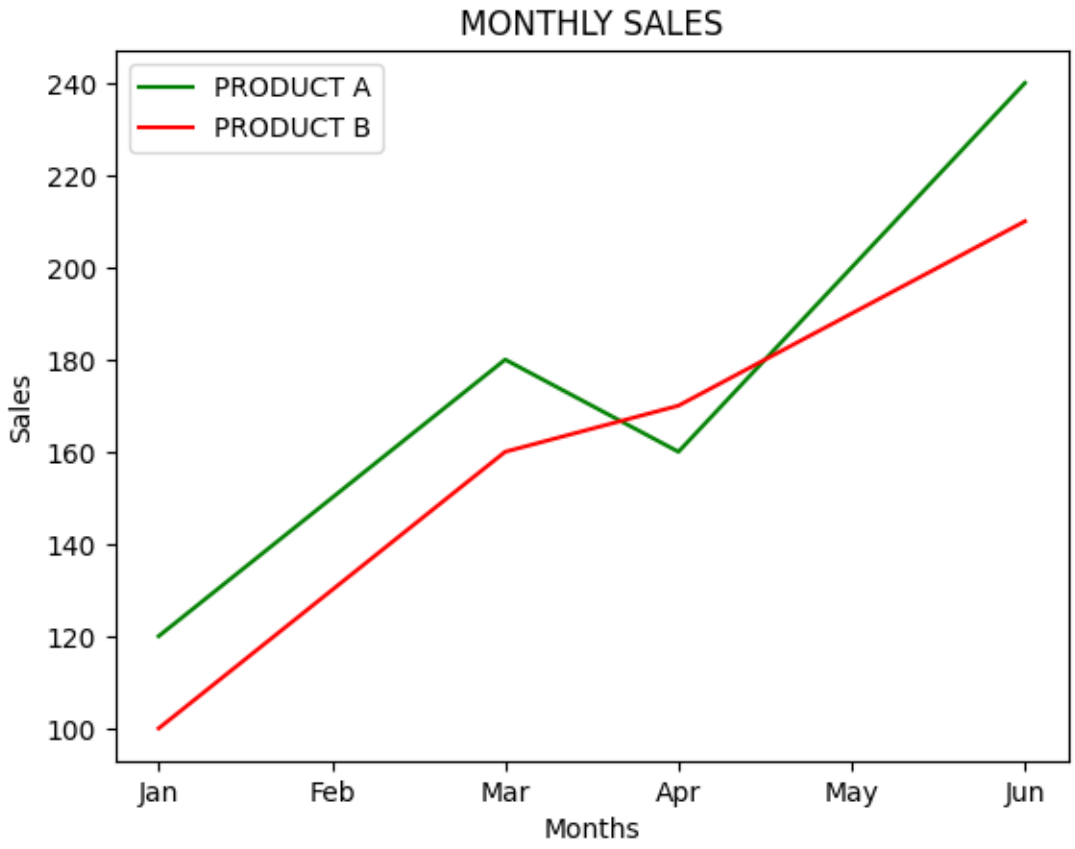
sales.xlabel("Months")

sales.ylabel("Sales")

sales.legend(loc="upper left")

sales.show()
```

OUTPUT:



RESULT:

Thus, the monthly sales of Product A and Product B were visualized successfully using a line plot. The graph shows the sales trends of both products from January to June.

Exp No:3.b Conduct an experiment to show data visualization using bar chart

AIM:

To conduct an experiment to show data visualization using bar chart

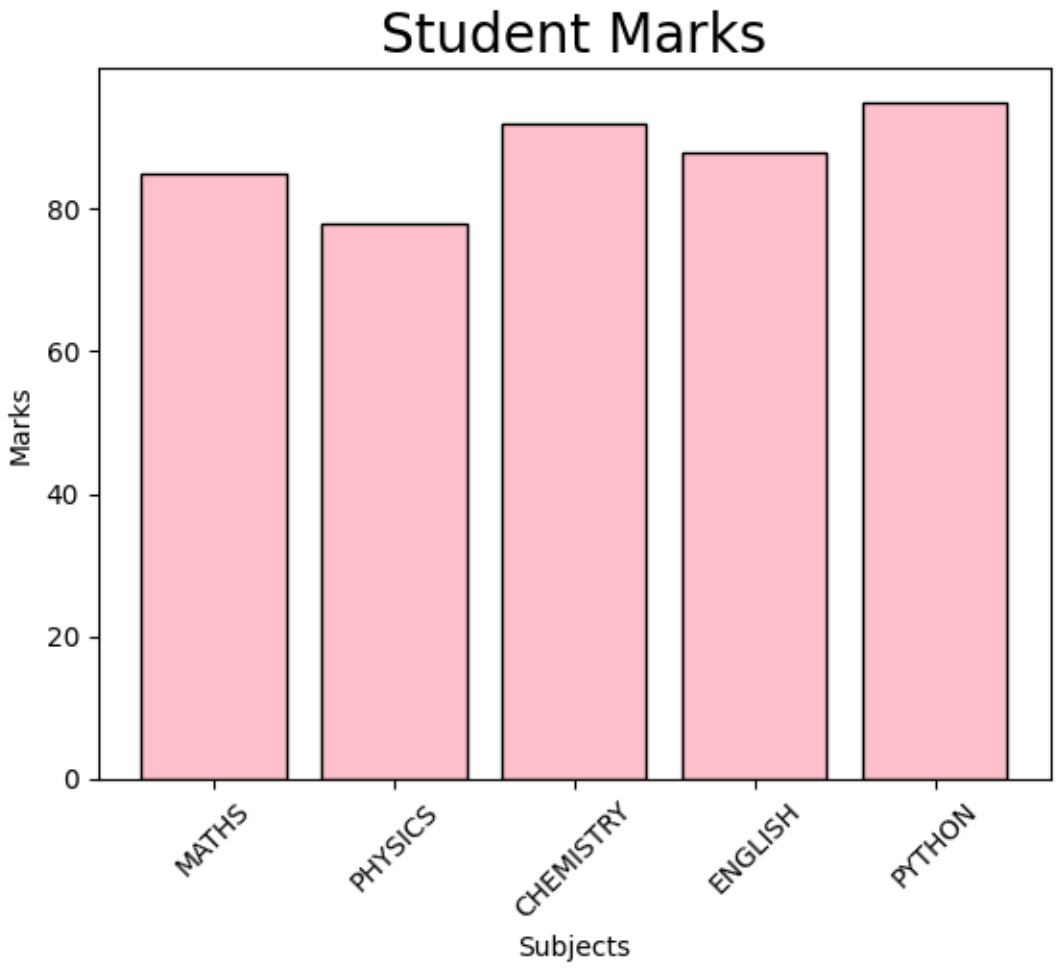
DESCRIPTION:

Take any sample data either through csv file or data fetched directly through code.

CODE:

```
import matplotlib.pyplot as marks
import numpy as np
Subjects = ['MATHS', 'PHYSICS', 'CHEMISTRY', 'ENGLISH', 'PYTHON']
xaxis = np.arange(len(Subjects))
Marks = [85, 78, 92, 88, 95]
marks.bar(Subjects, Marks,color="pink",edgecolor="black")
marks.xticks(xaxis, Subjects, rotation=45)
marks.xlabel("Subjects")
marks.ylabel("Marks")
marks.title("Student Marks", fontsize=20, color="black")
marks.show()
```

OUTPUT:



RESULT:

Thus, the student marks were visualized successfully using a bar chart. The chart compares marks across five subjects, with Python scoring the highest marks and Physics the lowest.

Exp No:3.c Conduct an experiment to show data visualization using pie chart

AIM:

To conduct an experiment to show data visualization using pie chart

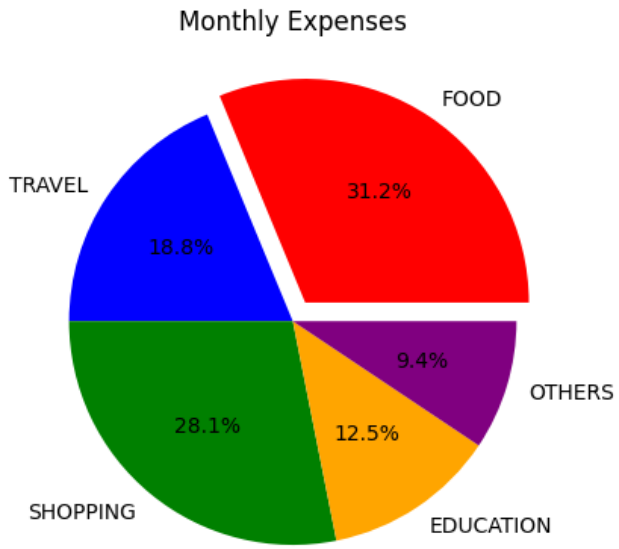
DESCRIPTION:

Take any sample data either through csv file or data fetched directly through code.

CODE:

```
import matplotlib.pyplot as expense
# Monthly expense data
labels = ['FOOD', 'TRAVEL', 'SHOPPING', 'EDUCATION', 'OTHERS']
amount = [5000, 3000, 4500, 2000, 1500]
colors = ['red', 'blue', 'green', 'orange', 'purple']
explode = (0.1, 0, 0, 0, 0)
# Plotting the pie chart
expense.pie(amount, labels=labels, colors=colors, explode=explode, autopct='%1.1f%%')
expense.title('Monthly Expenses')
expense.show()
```

OUTPUT:



RESULT:

Thus, the monthly expenses were visualized successfully using a pie chart. The chart shows the percentage distribution of expenses across five categories, with food accounting for the highest expenditure.